ENTREPRE-NEURSHIP GUIDE

Entrepreneurship Guide

A student team with a new technology often starts with a platform and looks for uses. This guide turns what we learned from AeroSense's entrepreneurship work into steps another team can follow, from a technology that can do many things to a customer with one problem. It rests on one project and on written feedback from mentors and reviewers, so treat it as a starting point and adapt it to yours.

1. Start from the customer's pain, not your technology

One mentor panel opened its review with a simple instruction: find the market's pain first, then check whether your own strength actually addresses it, and not the reverse. We had been doing it the other way around. Write the pain as a specific loss that a specific customer can name.

2. Choose one beachhead

A technology that can do many things is not the same claim as a customer with one expensive problem. We started with six crops and narrowed to one, with the others named as follow-on markets. Every extra application subtracts attention from the one story that people will believe: a real shipment and a real loss.

3. Size the market from the customer's loss

Our first market sizing came from the global market for the instrument we were building, which never connected back to what a customer would lose. We rebuilt it bottom-up from the opportunity itself, in three layers: the total market, the part we can serve in our region and sector, and the part we can realistically win first. Show the logic for each number, and mark any number you have not confirmed as pending.

4. Compete on an axis you can defend

A comparison is not neutral. We first compared ourselves with other electronic-nose products mainly on analysis time, which is where general-purpose laboratory instruments beat us. We rebuilt the comparison around the dimensions where we can defend our position and dropped time as the headline. An interview later showed that "cheaper than the lab instrument" was also the wrong axis. What we could support was three words: fast, on-site, and low operator dependence.

5. Know who pays

Ask who the buyer is, separately from who the user is. Individual farmers had a low willingness to pay and often did not want their product flagged. Associations, cooperatives and certifying bodies were the more realistic buyers. Split the business model into the layers you will actually sell, such as hardware, consumables, software and custom projects.

6. Check regulation and rights early

A long list of applications hides very different paths to market. A diagnostic device faces a much heavier approval path than a screening tool for agriculture, and we had presented them as equally near. We moved our first application to the one with the more accessible path and kept the other as a long-term lead. Ask early about licensing of any tool you depend on, about equity with your university, and about who owns the software.

7. Show why your team can build it

Reviewers ask why this group can build this. A slide of names and photos does not answer it. We replaced ours with a capability matrix that shows which skills the team has and which it still needs. Lead with who you do it for, not with everything the platform can do, or it reads as a science project.

8. Test the pain with real customers, and use reviewers as rounds

Every cycle before this one rested on an assumption we had not tested directly: that customers lose enough produce, often enough, to pay for early screening. Ask customers instead of reasoning about them. When a request for data gets a guarded reply, lower the ask to a short conversation about how they do the job today. Treat each mentor and reviewer round as an experiment: record what you changed after each, and report the questions that are still open.

Checklist

- ☐ We can name one customer and one specific loss.
- ☐ We chose one beachhead, and other markets are follow-on.
- ☐ Our market sizing starts from the customer's loss.
- ☐ We compare ourselves on axes we can defend.
- ☐ We know who pays, and it may not be the user.
- ☐ We know the regulatory path and who owns the key rights.
- ☐ Our team slide shows capabilities, not just names.
- ☐ We have asked real customers, and we list what is still open.

Our materials

- Our Entrepreneurship page: https://pinche567.github.io/iGEM-2026-wiki/entrepreneurship.html